

# Blowup profiles for the Okamoto, Sakajo and Wunsch family: exact stagnation point relations and the regularity exponent

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## Abstract

We study finite time blowup in the one parameter family of one dimensional models for the three dimensional Euler vorticity equation introduced by Okamoto, Sakajo and Wunsch, which interpolates between the Constantin, Lax and Majda model at  $a = 0$  and the De Gregorio model at  $a = 1$ . Starting from  $\omega_0 = \sin x$  on the circle we find finite time blowup for every  $a < 1$  with rate exponent one, and two structurally distinct mechanisms: for small  $a$  the peak narrows, while for  $a$  in a window around 0.8 the normalised spectrum freezes and the solution converges to a fixed profile whose amplitude alone diverges. In the second regime the profile satisfies the fixed point equation  $\mathcal{R}(f) = f$ , where  $\mathcal{R}$  is the same quadratic nonlinearity that drives the evolution. This equation admits no free scaling, so it determines the profile amplitude outright, and that amplitude is independently measurable from the simulation; the two agree to  $10^{-5}$ . The linearisation about the profile, taken in renormalised time, has exactly two eigenvalues outside the open left half plane, both forced by symmetry, so the profile is linearly stable and is selected by the dynamics. We show that the profile can only be singular where its own velocity vanishes, that the two such stagnation points are exactly  $\pi$  apart, and that the local exponent there is  $\mu = (c - 1)/(ac)$  with  $c$  the Hilbert transform of the profile at that point. At the stagnation point where the profile has a simple zero this closes in elementary form,  $c = 1/(1 - a)$ , verified to machine precision. The remaining stagnation point governs the spectral decay  $|\hat{f}_k| \sim k^{-\beta}$  with  $\beta = \mu + 1$ ; its value is genuinely global. Using the exact relation as a control variate we determine it to five digits or better and conclude that  $\beta$  varies continuously with  $a$  and is not equal to 3. We also give an exact first integral of the profile equation and a global constraint,  $\text{PV} \oint dy/U = 0$ , which we verify to  $10^{-12}$ .

## 1 Introduction

The question of whether solutions of the three dimensional incompressible Euler equations can develop singularities from smooth data remains open, and the same question for Navier and Stokes is one of the Clay Millennium Prize problems. A recurring strategy is to study one dimensional models which retain the competition between vortex stretching and transport while discarding the geometry.

Constantin, Lax and Majda [1] introduced the model

$$\omega_t = \omega \mathcal{H} \omega, \tag{1}$$

where  $\mathcal{H}$  is the Hilbert transform, and solved it in closed form. De Gregorio [2] restored the transport term, and Okamoto, Sakajo and Wunsch [3] embedded both in the one parameter family

$$\omega_t + a u \omega_x = \omega u_x, \quad u_x = \mathcal{H}\omega, \quad (2)$$

posed here on the circle with  $u$  of zero mean. The parameter  $a$  tunes the strength of transport against stretching. At  $a = 0$  transport is absent and (2) reduces to (1); at  $a = 1$  the two terms are in genuine competition, and this case is understood to be critical. Jia, Stewart and Šverák [4] studied the ground state of the De Gregorio model, Lei, Liu and Ren [5] proved global regularity on the circle for data of one sign, and Chen, Hou and Huang [6] proved finite time blowup on the line for data of low Hölder regularity. The broader programme of computer assisted singularity detection in fluid equations is exemplified by Luo and Hou [7], Elgindi [8] and Chen and Hou [9].

Two features of (2) make it a good testbed. Both endpoints of the family are known exactly, which makes every numerical statement checkable, and the solution structure is simple enough that the blowup profile can be computed directly rather than inferred from a diverging simulation.

**Relation to existing work.** Self similar blowup for this family is under active study and the present results must be read against it.

Okamoto, Sakajo and Wunsch [3] study global existence against blowup using a criterion in the spirit of Beale, Kato and Majda together with energy estimates, and do not consider self similar profiles. They do fit the spectrum in the combined form  $|\hat{\omega}_n| \sim C n^{-p} e^{-\delta n}$  and track both parameters in time, observing that  $\delta(t)$  decays exponentially, which they read as evidence of smoothness for all time, while  $p(t)$  decreases monotonically and, in their words, they are unable to see its asymptotic value from the numerical data. The exponent  $\beta$  of (13) is the analogue of that  $p$  for a frozen blowup profile, and (13) determines it in closed form given one local quantity.

Huang, Tong and Wei [11] construct infinitely many compactly supported self similar blowup solutions of the De Gregorio model on the real line, organised as eigenfunctions of a compact operator and as solutions of nonlinear eigenvalue problems, with profiles in  $H_0^1 \cap H^2$  and smooth in the interior of the support. Huang, Tong and Wang [16] study self similar blowup of the generalised model from data with derivative degeneracy, reporting one scale blowup with regular profiles for  $a > 0$  against a two scale scenario with singular outer profile for  $a \leq 0$ , and constructing an explicit family of singular profiles in the latter regime. Both use the standard self similar ansatz, whose profile equation  $(c_l x + a u) \omega_x = (c_\omega + u_x) \omega$  reduces to (7) at  $c_l = 0$ ,  $c_\omega = -1$ . We claim no novelty for that formulation.

Huang, Qin, Wang and Wei [12] prove that the family admits exact self similar blowup with interiorly smooth profiles for all  $a \leq 1$ , obtained as a fixed point of an  $a$  dependent nonlinear map, with characterisations of regularity, monotonicity and far field decay. Their profiles fail to be smooth only at the boundary of a compact support, whereas the profile here is  $C^{1,1}$  at an *interior* point. The two are consistent, and the reason is the domain: on the circle  $U$  has zero mean by construction, so it must change sign and interior stagnation points are forced, while on the line they are not. Their analogues of  $c_1$  and  $c_2$  are defined as integrals and bounded rather than evaluated, for instance  $2\eta/\pi \leq c(f) \leq 4/\pi$ , and no closed form appears.

Huang, Qin and Wang [13] construct blowups with two collapsing spatial scales on the line, one measuring the distance to the blowup point and one the shrinking bulk. Those profiles

narrow,  $c_l > 0$ , so they belong to the narrowing regime of Section 4 rather than the frozen one, and the blowup there occurs where  $\omega_0$  vanishes with  $\mathcal{H}\omega_0 > 0$  rather than at a zero of the velocity.

Guo and Jiu [15] study stability on the torus at  $a = 1$  near the excited state  $-\sin 2\theta$ , proving linear and nonlinear instability for a broad class of data. That is the steady state analogue, at the critical parameter, of the observation in Section 6 that the higher members of Proposition 4 are not generic attractors. It concerns equilibria rather than blowup profiles and takes no renormalised time limit.

Zheng [14] constructs exactly self similar  $C^\alpha$  blowup for the generalised De Gregorio equation at small  $|a\alpha|$ , again on the line and again narrowing, with rescaling factor  $(1 - t)^{1+\lambda(a)}$ .

A pattern runs through all of these. Every construction we are aware of works on the real line with a narrowing profile,  $c_l \neq 0$ . The circle behaves differently for a reason that is not a matter of taste: there  $U$  has zero mean by construction, so it must change sign, interior stagnation points are forced, and by Proposition 6 the profile cannot be smooth at them. That is the setting in which a frozen profile,  $c_l = 0$ , becomes the attractor and in which the local relations below have content. None of these references performs the local analysis at stagnation points which gives Propositions 6 and 8, nor the linear stability computation of Section 7. Two numerical coincidences with [16] invited closer comparison, and Remark 7 is what makes that comparison possible, since it puts both settings in one formula.

**The transition near  $a \approx 0.75$ .** [16] identifies  $a^{c,2} \approx 0.751$  as the value at which its self similar profile *loses* stability as  $a$  increases. The frozen profile studied here *gains* selection as  $a$  increases through roughly the same value. These are not the same object: their profiles carry  $c_l \neq 0$  and therefore narrow, which in the language of Section 4 places them in the narrowing regime, whereas a frozen profile requires  $c_l = 0$  exactly. It is tempting to read the two as complementary halves of one exchange of stability, in which the narrowing attractor gives way to the frozen one.

Table 2 tests that reading. The mechanism survives and the location does not.

The mechanism:  $c_l$  falls smoothly to zero rather than jumping, so the narrowing branch merges with the frozen one rather than being replaced by it, which is the right shape for an exchange.

The location:  $c_l$  is not zero at  $a = 0.751$ . It measures  $0.004536 \pm 0.000543$  there, eight standard errors from zero, and nine times the conservative  $5 \times 10^{-4}$  noise floor. Whatever happens at 0.751 on the line, the narrowing branch on the circle is still narrowing there. The merger itself is bracketed rather than determined, since the exponent and the root trade off against each other in a fit of  $c_l = A(a_c - a)^p$ : the best  $a_c$  runs from 0.795 at  $p = 2$  to 0.844 at  $p = 4.5$ , with the fit quality improving monotonically in  $p$ , so all that follows is  $a_c \gtrsim 0.795$ .

We therefore withdraw the coincidence as refuted rather than unsupported. This is not a contradiction of [16]. A branch merger and a loss of stability are different events which need not occur together, and their  $a^{c,2}$  is a stability boundary for a profile on the line while  $a_c$  here is where a scaling exponent vanishes on the circle. The conclusion is only that the two numbers are not the same number, and that the resemblance which prompted the comparison was an artefact of a linear extrapolation through convex data, the same trap as in Section 9.

**The constant  $1/(1 - a)$ .** Theorem 2.6 of [16] gives, for  $a < 0$ , an exact solution  $\bar{\Omega}_a(X) = -\sin(\pi\mu) \mathbf{1}_{X>1} |1 - X|^{-\mu}$  with  $\mu = 1/(1 - a)$  and scaling factors  $c_{l,a} = 1 - a$ ,  $c_{\omega,a} = -1$ . The

resemblance to Proposition 8 is superficial. Their  $\mu$  is an exponent and equals  $1/c_{l,a}$  in their normalisation; ours is a value of  $\mathcal{H}f$  at a point where it is finite. Equation (12) does not even apply at their singular point, since  $\mathcal{H}\bar{\Omega} = 2/(1 - X^2)$  diverges there and the advection velocity does not have a simple zero. What the two share is the algebraic origin of the factor:  $(1 - a)$  is what appears whenever the stretching coefficient 1 is balanced against the advection coefficient  $a$ , entering our case through  $h(1 - a) = 1 + c_l$  and theirs through the scaling normalisation. Neither statement implies the other.

Our contributions are as follows.

1. We separate two blowup mechanisms within the family and give a translation invariant diagnostic that distinguishes them (Section 4).
2. We formulate the blowup profile equation as a fixed point of the evolution nonlinearity, observe that it fixes the profile amplitude with no free scaling, and verify the resulting prediction against simulation to  $10^{-5}$  (Section 5).
3. We compute the spectrum of the linearisation in renormalised time and identify the two symmetry forced eigenvalues exactly, one of which is provable in closed form (Section 7).
4. We prove that singularities of the profile occur only at stagnation points of its velocity, derive the local exponent, and show that one of the two stagnation points admits the elementary closed form  $c = 1/(1 - a)$  (Section 8).
5. We determine the remaining, global constant to five digits using the exact relation as a control variate, and conclude that the spectral decay exponent varies continuously with  $a$  (Section 9).
6. We give a first integral of the profile equation and a global constraint which we verify numerically (Section 10).

Throughout, the initial datum is  $\omega_0 = \sin x$ . This is not a generic choice, and Section 12 records what that costs.

## 2 The model and its exactly known endpoints

Write  $\omega$  for the vorticity on the circle  $x \in [0, 2\pi)$  and let  $\mathcal{H}$  denote the Hilbert transform, acting on Fourier modes as  $\widehat{\mathcal{H}\omega_k} = -i \operatorname{sgn}(k) \hat{\omega}_k$ . The velocity is recovered from  $u_x = \mathcal{H}\omega$  with  $u$  of zero mean, so  $\hat{u}_k = -\hat{\omega}_k/|k|$  for  $k \neq 0$ .

**The case  $a = 0$ .** Setting  $z = \mathcal{H}\omega + i\omega$  converts (1) into the Riccati equation  $z_t = z^2/2$ , whence

$$\omega(x, t) = \frac{4\omega_0(x)}{(2 - t\mathcal{H}\omega_0(x))^2 + t^2\omega_0(x)^2}. \quad (3)$$

For  $\omega_0 = \sin x$  the denominator collapses to  $4 + 4t \cos x + t^2$  and

$$\|\omega(t)\|_\infty = \frac{4}{4 - t^2}, \quad (4)$$

so the blowup time is  $T = 2$  and the rate exponent is exactly one.

**The case  $a = 1$ .** For  $\omega_0 = \sin x$  one has  $\mathcal{H}\omega_0 = -\cos x$  and  $u = -\sin x$ , so  $u\omega_x = \omega u_x$  pointwise and  $\sin x$  is a steady state. Hence  $T = \infty$ .

The family therefore runs between a case whose blowup time is exactly 2 and a case whose blowup time is exactly infinite, and any numerical scheme can be calibrated against both.

### 3 Numerical method

Space is treated pseudospectrally: products are formed on an equispaced grid of  $N$  points, dealiased by the two thirds rule, with the Nyquist mode set to zero. Time stepping is classical fourth order Runge and Kutta, with the step governed by two constraints. Accuracy near blowup requires  $\Delta t = \text{cfl} / \max(\|u_x\|_\infty, \|\omega\|_\infty)$ , since the stretching term drives the growth. Stability requires  $\Delta t < c_{\text{adv}} / (k_{\text{cut}} |a| \|\omega\|_\infty)$ , because the transport term has eigenvalues on the imaginary axis out to  $i a u k_{\text{cut}}$  and the method is stable there only to 2.828.

Two diagnostics require care. First, the supremum norm is computed from the spectral interpolant rather than the grid samples: the grid maximum errs by  $O(\Delta x^2 f'')$ , and since the peak narrows like  $1/\|\omega\|_\infty$  near blowup, that error grows precisely when it matters. We locate the peak by Newton iteration on  $f'(x) = 0$  against the interpolant, at  $O(N)$  cost for a single point. Second, spectral decay rates are measured over a band bounded away from both ends: the low wavenumber end is contaminated by the analytic factor multiplying the singular behaviour, and the high wavenumber end by the dealiasing mask. Section 9 quantifies both.

The implementation passes nineteen checks against closed forms and conserved quantities, including (3) and (4) to  $2 \times 10^{-11}$ , conservation of  $\int \omega$  to  $10^{-16}$  for  $a \in \{0, \frac{1}{2}, 1, -\frac{1}{2}\}$ , spatial convergence following the predicted analyticity strip across four orders of magnitude, and fourth order temporal convergence with measured ratios 16.65, 16.28, 16.16, 16.07.

### 4 Blowup times and two distinct regimes

From  $\omega_0 = \sin x$  we observe finite time blowup for every  $a < 1$ , with rate exponent one throughout. Table 1 collects the blowup times, estimated by fitting  $\omega / (d\omega/dt) = (T - t)/p$ , a straight line in  $t$  with slope  $-1/p$  and root  $T$ .

| $a$ | 0         | 0.2    | 0.4    | 0.5    | 0.6    | 0.7    | 0.8    | 0.9    | 1                |
|-----|-----------|--------|--------|--------|--------|--------|--------|--------|------------------|
| $T$ | 2 (exact) | 2.3101 | 2.7858 | 3.1416 | 3.6500 | 4.4641 | 6.0719 | 10.757 | $\infty$ (exact) |
| $p$ | 1 (exact) | 0.997  | 1.000  | 1.001  | 1.001  | 0.999  | 1.000  | 1.049  | n/a              |

Table 1: Blowup time and rate exponent from  $\omega_0 = \sin x$ . No transition to global existence appears below  $a = 1$ ; instead  $T$  diverges as  $a \rightarrow 1$ .

**Remark 1.** The value at  $a = 1/2$  agrees with  $\pi$  to  $1.2 \times 10^{-11}$  and continues to do so under refinement across an eightfold range in  $N$ . We have no explanation for this and record it as an observation.

The blowups are not all alike. Comparing normalised spectra  $|\hat{\omega}_k| / \max_k |\hat{\omega}_k|$  at successive decades of amplitude growth gives a diagnostic which is invariant under translation and assumes nothing about the profile shape. At  $a = 0.8$  the curves converge: successive deviations from the

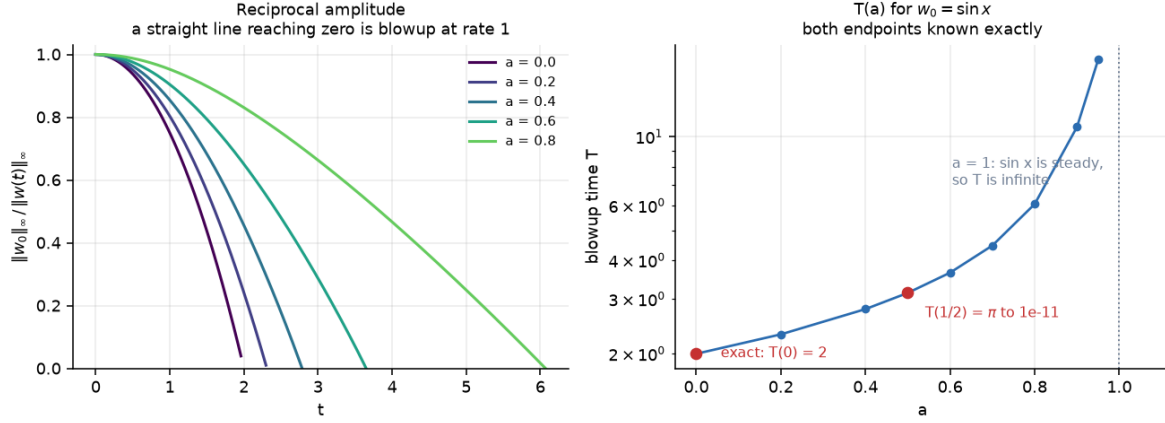


Figure 1: Left: reciprocal amplitude  $\|w_0\|_\infty / \|w(t)\|_\infty$  for several values of  $a$ . A straight line reaching zero is blowup with rate exponent one. Right: the blowup time  $T(a)$  from  $w_0 = \sin x$ , on a logarithmic scale. Both endpoints are known exactly,  $T(0) = 2$  and  $T(1) = \infty$ , and the curve runs between them without any transition to global existence at intermediate  $a$ . The agreement of  $T(1/2)$  with  $\pi$  is marked but unexplained.

first snapshot are  $3.30 \times 10^{-3}$ ,  $3.68 \times 10^{-3}$ ,  $3.76 \times 10^{-3}$ ,  $3.78 \times 10^{-3}$  across four decades of growth, the peak location locks to  $x = 2.16699$ , and the spectral tail remains at  $2.7 \times 10^{-17}$  throughout, so a grid of 4096 points carries an amplitude of  $10^6$  without strain. At  $a = 0.4$  the same comparison moves by 0.72 over a single decade and resolution is exhausted by  $\|w\|_\infty = 5.8 \times 10^3$ .

We refer to the first behaviour as a *frozen* profile and the second as *narrowing*, illustrated in Figure 2. Only the frozen case is analysed below.

The distinction is quantitative. In the general self similar ansatz of Remark 7, the peak width scales like  $(T - t)^{c_l}$  while the amplitude grows like  $(T - t)^{-1}$ , so width  $\sim$  amplitude $^{-c_l}$  and  $c_l$  is minus the slope of one against the other on logarithmic axes. A frozen profile is exactly  $c_l = 0$ . Measuring the width as the half width at half maximum around the peak, which is robust whether the spectrum is exponential or algebraic, gives Table 2. The method is calibrated at  $a = 0$ , where the exact solution (3) has width proportional to  $(2 - t)$  and amplitude  $4/(4 - t^2)$ , so  $c_l = 1$ ; the measurement returns 1.048.

|           |                      |                      |                      |                      |                      |
|-----------|----------------------|----------------------|----------------------|----------------------|----------------------|
| $a$       | 0                    | 0.2                  | 0.4                  | 0.5                  | 0.6                  |
| $c_l$     | 1.058                | 0.748                | 0.512                | 0.325                | 0.179                |
| $a$       | 0.70                 | 0.74                 | 0.75                 | 0.77                 | 0.80                 |
| $c_l$     | 0.032321             | 0.007273             | 0.004536             | 0.001294             | 0.000398             |
| std. err. | $1.5 \times 10^{-3}$ | $7.5 \times 10^{-4}$ | $5.4 \times 10^{-4}$ | $3.0 \times 10^{-4}$ | $1.2 \times 10^{-4}$ |

Table 2: The spatial scaling exponent measured from the dynamics. The exact value at  $a = 0$  is 1. The upper block is resolution limited and quoted to two figures; the lower block is fitted over three decades of amplitude, to  $10^6$ . Values at  $a \geq 0.78$  are omitted because they are not monotone in  $a$  at the  $5 \times 10^{-4}$  level, which is the true noise floor rather than the smaller figure the regression reports.

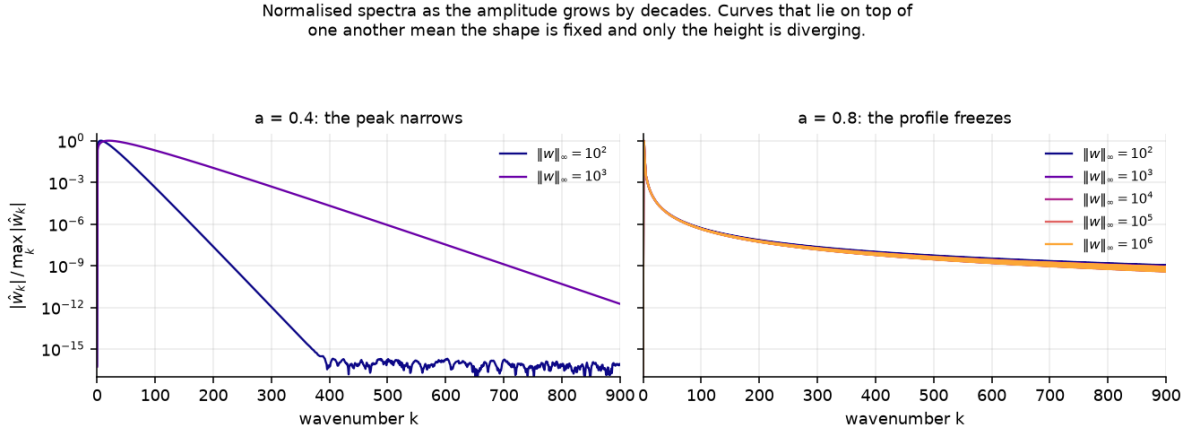


Figure 2: Normalised spectra  $|\hat{w}_k| / \max_k |\hat{w}_k|$  at successive decades of amplitude growth. Taking the modulus removes the translation, so no alignment of the peaks is required and the comparison assumes nothing about the profile shape. Left,  $a = 0.4$ : the curves spread steadily as the peak narrows, and resolution is exhausted by  $\|w\|_\infty = 5.8 \times 10^3$ . Right,  $a = 0.8$ : five curves spanning four decades of growth lie on top of one another, so the shape is fixed and only the height diverges.

**Remark 2** (One signed data). Lei, Liu and Ren [5] proved global regularity on the circle for data of one sign at  $a = 1$ . The mechanism appears to reach well below that. From  $\omega_0 = 1 - \cos x$ , integrating to  $t = 200$ , the amplitude remains bounded at 2.72, 2.40 and 2.17 for  $a = 0.7, 0.8$  and  $0.9$ , while for  $a = 0.6$  the solution reaches  $10^6$  by  $t = 133$  and for  $a = 0.5$  it exhausts resolution at  $9.6 \times 10^3$ . The threshold lies between 0.6 and 0.7. We record this as an observation; it says nothing about the frozen profile, which is reached from sign changing data.

## 5 The profile equation

Substituting the ansatz

$$\omega(x, t) = \frac{f(x - x_p)}{T - t} \quad (5)$$

into (2) gives  $f/(T - t)^2$  on the left and  $\mathcal{R}(f)/(T - t)^2$  on the right, where

$$\mathcal{R}(f) = f \mathcal{H}f - a U f', \quad U' = \mathcal{H}f, \quad \int U = 0 \quad (6)$$

is the same quadratic nonlinearity that drives the evolution. The profile equation is therefore the fixed point problem

$$\boxed{\mathcal{R}(f) = f.} \quad (7)$$

**Proposition 3** (No free scaling). *If  $f$  solves (7) and  $c \neq 1$ , then  $cf$  does not.*

*Proof.*  $\mathcal{R}$  is quadratic, so  $\mathcal{R}(cf) = c^2 \mathcal{R}(f) = c^2 f$ , which equals  $cf$  only when  $c^2 f = cf$ , that is  $c = 1$ .  $\square$



Proposition 3 is what makes (7) useful rather than merely descriptive: unlike a linear eigenvalue problem it fixes the amplitude of  $f$  outright. Since (5) gives  $\|\omega(t)\|_\infty = \|f\|_\infty / (T - t)$ , we obtain the falsifiable prediction

$$\frac{d}{dt} \frac{1}{\|\omega\|_\infty} = -\frac{1}{\|f\|_\infty}. \quad (8)$$

The left side is measured by time marching the partial differential equation; the right side by Newton's method on an algebraic fixed point. The two share nothing but the value of  $a$ .

We solve (7) by Newton iteration, assembling the Jacobian one column at a time through the same masked transforms used to evaluate  $\mathcal{R}$ , and regularising the translation mode by a rank one shift along  $f'$ . At  $a = 0.8$  the profile taken directly from the simulation satisfies (7) to  $1.3 \times 10^{-5}$  before Newton is applied; the iteration then converges quadratically, with residuals  $1.35 \times 10^{-5}$ ,  $4.07 \times 10^{-11}$ ,  $1.77 \times 10^{-12}$ . Table 3 compares the two sides of (8).

| $N$  | simulation, via (8) | profile equation | agreement            |
|------|---------------------|------------------|----------------------|
| 2048 | 4.148459148         | 4.148481857      | $5.5 \times 10^{-6}$ |
| 4096 | 4.154779433         | 4.154737187      | $1.0 \times 10^{-5}$ |

Table 3:  $\|f\|_\infty$  at  $a = 0.8$ , computed two independent ways.

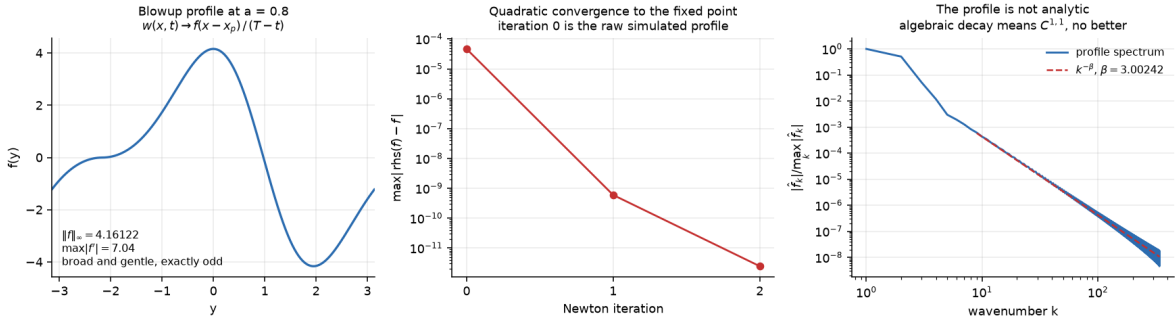


Figure 3: The blowup profile at  $a = 0.8$ . Left:  $f(y)$ , broad and gentle, with  $\|f\|_\infty = 4.16$  against  $\max|f'| = 7.04$ , and exactly odd about each stagnation point. Centre: the Newton residual for (7), starting from the profile taken directly out of the simulation, which already satisfies the equation to  $1.3 \times 10^{-5}$ ; convergence is quadratic. Right: the profile spectrum on logarithmic axes against  $k^{-\beta}$  with  $\beta = 3.00242$  from (13), the exponent predicted in Section 9 rather than fitted to this curve. The decay is algebraic, not exponential, so the profile is not analytic.

The shared value drifts by  $1.5 \times 10^{-3}$  between the two resolutions, which is not Newton finding different branches: re-solving at  $N = 4096$  from the interpolated  $N = 2048$  solution returns the same answer, and mapping it back down reproduces the  $N = 2048$  value to  $2 \times 10^{-12}$ . Each grid has one solution and the simulation finds that grid's solution. The slow convergence is explained in Section 8: the profile is only  $C^{1,1}$ .

**Proposition 4** (Dilation). *If  $f$  solves (7) then so does  $f(n \cdot)$  for every positive integer  $n$ , with the same  $\|f\|_\infty$ .*



*Proof.* The Hilbert transform commutes with integer dilation on the circle, so  $\mathcal{H}[f(n \cdot)](x) = (\mathcal{H}f)(nx)$ . Since the velocity is recovered by integrating  $U' = \mathcal{H}f$ , the velocity of  $f(n \cdot)$  is  $U(nx)/n$ , while  $\frac{d}{dx}f(nx) = nf'(nx)$ . The two factors of  $n$  cancel in the advection term, so

$$\mathcal{R}(f(n \cdot))(x) = f(nx)(\mathcal{H}f)(nx) - a \frac{U(nx)}{n} nf'(nx) = [f \mathcal{H}f - a U f'](nx) = f(nx). \quad \square$$

So profiles come in families. The  $n$ th member carries  $2n$  stagnation points spaced  $\pi/n$ , with exponents alternating between the two values of the base member, and  $\|f\|_\infty$ ,  $c_1$ ,  $c_2$  and  $\beta$  are all unchanged. Numerically,  $f(n \cdot)$  solves (7) to  $10^{-11}$  for  $n \leq 4$ , against  $2 \times 10^{-12}$  for the base profile itself.

**Genericity.** Every result here starts from  $\omega_0 = \sin x$ , which is the ground state of the  $a = 1$  model and therefore not a generic datum. Proposition 5 and the stability computation of Section 7 predict that any datum in the basin converges to the same profile, and this is testable independently of them. At  $a = 0.8$  and  $N = 2048$ :

| datum                                | $\ f\ _\infty$ | $c_1$    | $c_2$      | $\beta$  |
|--------------------------------------|----------------|----------|------------|----------|
| $\sin x$                             | 4.1484819      | 4.996663 | -1.6580199 | 3.003911 |
| $\sin x + 0.3 \sin 2x$               | 4.1484819      | 4.996663 | -1.6580199 | 3.003911 |
| $\sin x + 0.5 \cos 2x - 0.2 \sin 3x$ | 4.1484819      | 4.996663 | -1.6580199 | 3.003911 |
| $\sin 3x + 0.4 \sin 4x$              | 4.1612203      | 5.006926 | -1.668103  | 2.999354 |

Table 4: The same profile is reached from every datum tested. The first three agree to seven digits. The fourth lands on the  $n = 2$  member of Proposition 4, which on a grid of 2048 points is represented as accurately as the base member on 1024 points, and indeed reproduces the  $N = 1024$  values of Table 6 exactly.

**Selection.** Solutions of (7) exist over a wider range of  $a$  than the dynamics selects. Starting Newton from each simulation's own profile and recording the residual *before* the iteration begins gives an order parameter: it reads 0.33 to 0.56 for  $a$  between 0.3 and 0.6, then 0.19,  $5.0 \times 10^{-2}$ ,  $2.6 \times 10^{-3}$ ,  $1.3 \times 10^{-5}$  and  $5.5 \times 10^{-5}$  at  $a = 0.65, 0.70, 0.75, 0.80, 0.85$ . The frozen regime is therefore roughly  $0.75 \leq a \leq 0.85$  at the amplitudes reached, degrading smoothly rather than switching off.

## 6 The basin, and a third outcome

Since  $\mathcal{R}$  is quadratic, replacing  $\omega_0$  by  $\lambda\omega_0$  rescales time and changes nothing else, so the basin is scale invariant and only shape matters. Combined with Proposition 4, the question is which member a datum reaches.

**Member one is the generic attractor.** Sixteen random data, modes 1 to 8 with amplitudes falling like  $1/k$  and random phases, carrying between 4 and 10 sign changes, all reach member one with  $\|f\|_\infty = 4.148482$  in every case. The number of sign changes does not select the member. Higher members are reached only from data with the matching exact symmetry:  $\sin(kx)$  gives member  $k$  for  $k \leq 4$ , and since  $\sin(kx)$  is  $2\pi/k$  periodic it remains in that symmetric subspace, while any generic perturbation returns the solution to member one.

**A topological boundary.** For  $\omega_0 = \sin x + m$ , which changes sign for  $m < 1$  and is one signed at  $m = 1$ , blowup occurs for every  $m < 1$  with  $\|f\|_\infty = 4.148482$  throughout and  $T$  growing as 6.07, 10.07, 15.52, 20.28, 25.88, 43.81 at  $m = 0, 0.5, 0.8, 0.9, 0.95, 0.99$ , and ceases at  $m = 1$  exactly, where the sign change degenerates into a touching zero. We fitted the divergence of  $T$  and report no law: neither  $-\log(1 - m)$  nor  $(1 - m)^{-p}$  has stable constants over this range.

**A third outcome.** Sign change is not sufficient. For  $\omega_0 = \sin x + c \sin 2x$  the two members compete, and between them lies a window in which the solution does neither:

| $c$     | 0.46     | 0.475    | 0.48     | 0.50  | 0.52  | 0.56     | 0.58     |
|---------|----------|----------|----------|-------|-------|----------|----------|
| outcome | member 1 | member 1 | member 1 | decay | decay | member 2 | member 2 |
| $T$     | 61.4     | 90.4     | 108.2    | -     | -     | 54.9     | 44.0     |

Table 5:  $T$  diverges on approach from both sides. Inside the window the amplitude falls monotonically: at  $c = 0.5$  it reaches  $7.4 \times 10^{-3}$  by  $t = 400$  with  $\|\omega\|_\infty t$  settling on 3.09, 3.03, 2.99, 2.97, so the solution decays like  $1/t$ .

The window is open rather than a separatrix, since both  $c = 0.50$  and  $c = 0.52$  fail to blow up while 0.48 and 0.56 do, so the model admits global decaying solutions from an open set of sign changing data. The window is narrow, roughly  $0.49 < c < 0.55$ .

Decay like  $1/t$  invites the guess that the interior is self similar. If  $\omega \sim g/t$  then  $R(g) = -g$ , and since  $\mathcal{R}$  is quadratic  $\mathcal{R}(-g) = \mathcal{R}(g)$ , so  $h = -g$  would solve (7): the attractor would be a profile entered with the opposite sign, of amplitude 2.95, matching neither member. Writing  $s = \log t$  and  $W = t\omega$  gives  $W_s = \mathcal{R}(W) + W$ , whose linearisation is  $-J$ , so such a profile would have to be strongly unstable as a blowup profile.

It is not so. Two tests refute it. Newton started from  $-t\omega$  at  $t = 400$  returns member 1, with  $\|f\|_\infty = 4.1484819$  and its constants and spectrum; the guess carried an eight percent residual and was never near a solution. And integrating to  $t = 2000$  separates the amplitude from the shape: the decay exponent converges to 1 and  $\|\omega\|_\infty t$  to 2.950, while the normalised spectrum drifts by  $4.7 \times 10^{-3}$ ,  $1.9 \times 10^{-2}$ ,  $4.6 \times 10^{-2}$  at  $t = 400, 1400, 2000$  and the residual of  $-t\omega$  grows from  $6 \times 10^{-2}$  to 4.9 with its sup norm unchanged. Fine structure is accumulating at high wavenumber rather than a profile settling.

The decay is therefore asymptotically  $1/t$  in amplitude and not self similar in shape. What organises the interior of the window is not a fixed profile and we do not identify it. The same mechanism accounts for  $c = 0.52$  exhausting resolution at  $t = 42$  while still decaying.

## 7 Linear stability in renormalised time

Set  $s = -\log(T - t)$  and  $W(y, s) = (T - t)\omega$ . Then  $\omega = e^s W$  and  $ds/dt = e^s$ , so  $\omega_t = e^{2s}(W + W_s)$ , while  $\mathcal{R}$  is quadratic and gives  $\mathcal{R}(\omega) = e^{2s}\mathcal{R}(W)$ . The factors cancel and

$$W_s = \mathcal{R}(W) - W. \quad (9)$$

The profile is exactly a fixed point of (9), and the eigenvalues of  $J = D[\mathcal{R}(\cdot) - \cdot](f)$  are growth rates in renormalised time.

**Proposition 5** (Forced eigenvalues).  $Jf = f$  and  $Jf' = 0$ .

*Proof.* Since  $\mathcal{R}$  is quadratic, its derivative satisfies  $D\mathcal{R}(f)f = 2\mathcal{R}(f) = 2f$ , so  $Jf = D\mathcal{R}(f)f - f = f$ . The second identity is translation invariance:  $\mathcal{R}$  commutes with shifts, so differentiating  $\mathcal{R}(f(\cdot + \epsilon)) = f(\cdot + \epsilon)$  at  $\epsilon = 0$  gives  $D\mathcal{R}(f)f' = f'$  and hence  $Jf' = 0$ .  $\square$

The eigenvalue 1 is not an instability. It reflects the freedom to shift the blowup time: replacing  $T$  by  $T + \epsilon$  gives  $W = f\tau/(\tau + \epsilon)$  with  $\tau = T - t$ , which to first order is  $f - \epsilon e^s f$  and grows like  $e^s$ . The eigenvalue 0 is translation. Numerically,  $\|Jf - f\|_\infty / \|f\|_\infty = 1.3 \times 10^{-12}$  and  $\|Jf'\|_\infty / \|f'\|_\infty = 4.0 \times 10^{-11}$ , so Proposition 5 also serves as a sharp check on the assembled Jacobian.

Every remaining eigenvalue at  $a = 0.8$  lies in the open left half plane, at  $N = 512, 1024$  and  $2048$  alike. The profile is therefore linearly stable up to the two symmetry directions, which is the precise sense in which the dynamics selects it, and is consistent with a simulation from  $\sin x$  reaching it without tuning. The same test applied to profiles taken from simulations at  $a = 0.65, 0.70, 0.75, 0.80$  returns stability in every case.

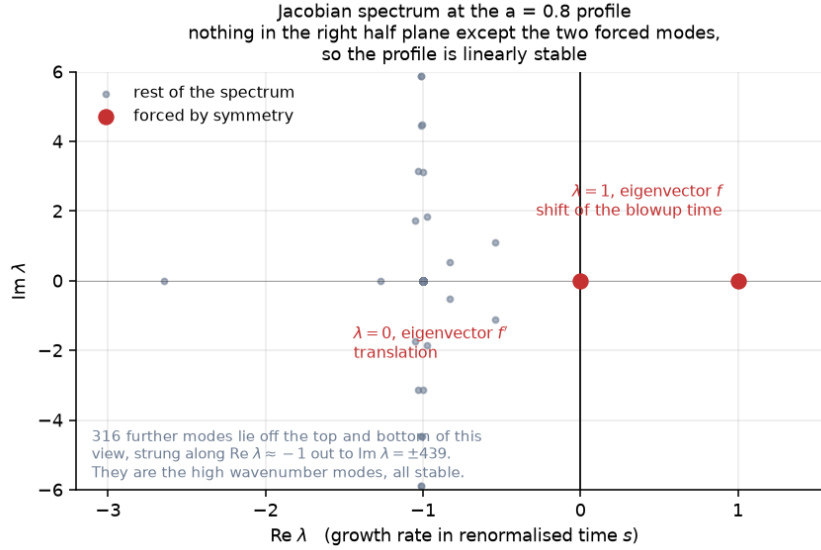


Figure 4: Spectrum of the linearisation (9) about the  $a = 0.8$  profile, so that the abscissa is a growth rate in renormalised time  $s$ . The two marked eigenvalues are forced by symmetry, at 1 with eigenvector  $f$  (freedom to shift the blowup time) and at 0 with eigenvector  $f'$  (translation), both verified against Proposition 5 to  $10^{-11}$  or better. Nothing else lies in the closed right half plane. The modes omitted from the view are the high wavenumber ones, strung along  $\text{Re } \lambda \approx -1$  and stable.

We note two limitations. The leading stable eigenvalue does not converge in  $N$ , taking values  $-0.54, -0.146, -0.41$  at  $N = 512, 1024, 2048$ , so the rate of approach to the profile is not determined here, only its sign. At  $a = 0.85$  and  $N = 1024$  the Newton solve lands on a different branch, with  $\|f\|_\infty = 6.249$  and an unstable eigenvalue at  $+2.077$ , whereas at  $N = 2048$  the same value of  $a$  yields  $\|f\|_\infty = 5.668$  matching its simulation to  $8 \times 10^{-6}$ . Continuation in  $a$  jumps branches outright. The branch structure is real and we have not mapped it.

## 8 Stagnation points and the local exponent

Rearranging (7) as  $aUf' = f(\mathcal{H}f - 1)$  gives

$$\frac{f'}{f} = \frac{\mathcal{H}f - 1}{aU}, \quad (10)$$

so  $f$  can only be singular where  $U$  vanishes.

**Proposition 6** (Local exponent). *Let  $U(y_*) = 0$  with  $U'(y_*) = c \neq 0$ . Then  $f(y_*) = 0$ , and  $f \sim C(y - y_*)^\mu$  with*

$$\mu = \frac{c - 1}{ac}. \quad (11)$$

*Proof.* Evaluating  $aUf' = f(\mathcal{H}f - 1)$  at  $y_*$  gives  $0 = f(y_*)(c - 1)$ , so  $f(y_*) = 0$  unless  $c = 1$ . Expanding  $U(y) = c(y - y_*) + O((y - y_*)^2)$  and using  $U' = \mathcal{H}f$ , so that  $\mathcal{H}f(y_*) = c$ , equation (10) becomes  $f'/f = \mu/(y - y_*) + O(1)$  with  $\mu$  as stated, and integrating gives the claim.  $\square$

**Remark 7** (General self similar form). The same computation applies to the full self similar ansatz  $\omega = (T - t)^{c_\omega} \Omega(x/(T - t)^{c_l})$ , whose profile equation is  $(c_l X + a\bar{U})\bar{\Omega}_X = (c_\omega + \bar{U}_X)\bar{\Omega}$ . A stagnation point is now a zero of the total advection velocity  $c_l X + a\bar{U}$  rather than of  $\bar{U}$  alone, and writing  $h = \mathcal{H}\bar{\Omega}(X_*)$  the exponent is

$$\nu = \frac{c_\omega + h}{c_l + ah}. \quad (12)$$

Proposition 6 is the case  $c_l = 0$ ,  $c_\omega = -1$ , which is what a frozen profile requires: no spatial rescaling. Likewise a simple zero forces  $h = (1 + c_l)/(1 - a)$ , of which Proposition 8 is again the  $c_l = 0$  case.

Since  $|y|^\mu$  has Fourier coefficients decaying like  $k^{-(\mu+1)}$ , the spectral decay exponent of the profile is

$$\beta = 1 + \frac{c - 1}{ac}. \quad (13)$$

One local number therefore predicts the entire spectral decay rate.

**Proposition 8** (Exact value at a simple zero). *If in addition  $f$  has a simple zero at  $y_*$ , that is  $f'(y_*) \neq 0$ , then*

$$c = \frac{1}{1 - a}. \quad (14)$$

*Proof.* Write  $f \sim A(y - y_*)$  with  $A = f'(y_*) \neq 0$ ,  $U \sim c(y - y_*)$  and  $\mathcal{H}f \sim c$ . Matching the coefficient of  $(y - y_*)$  in  $aUf' = f(\mathcal{H}f - 1)$  gives  $acA = A(c - 1)$ , and dividing by  $A$  yields  $ac = c - 1$ , that is  $c = 1/(1 - a)$ .  $\square$

Equivalently  $\mu = 1$  at such a point. Numerically, at  $N = 4096$ , the measured deviation of  $\mu$  from 1 is  $-1.7 \times 10^{-15}$ ,  $-4.6 \times 10^{-14}$ ,  $-2.0 \times 10^{-10}$  and  $3.5 \times 10^{-8}$  at  $a = 0.70, 0.72, 0.74, 0.76$ . The growth of this deviation at larger  $a$  tracks the profile's own accuracy, which degrades as the second exponent falls toward one.

**Two stagnation points, exactly  $\pi$  apart.** Numerically  $U$  has exactly two zeros, shown with the profile in Figure 5, and their separation is  $\pi$  to within  $9 \times 10^{-16}$  for every  $a$  tested. This is forced. If  $U(y_1) = 0$  and  $U(y_1 + \pi) = 0$ , then expanding both conditions in Fourier modes and adding annihilates the odd modes while subtracting annihilates the even ones, so the even mode part and the odd mode part of  $U$  must each vanish at  $y_1$  separately. That is two conditions where a generic zero satisfies one. Direct measurement at  $a = 0.8$  gives  $U_{\text{even}}$  and  $U_{\text{odd}}$  of order  $10^{-15}$  at both points, against  $\|U\|_{\infty} = 3.48$ . We emphasise that this is not the statement that  $f$  has only odd modes, which is false:  $|\hat{f}_2| = 3315$  against  $|\hat{f}_1| = 6520$ .

**Regularity of the profile.** At  $a = 0.8$  the second stagnation point has  $\mu \approx 2$ , and the general solution of (10) near a regular singular point permits different constants on the two sides. We find  $f''$  bounded and exactly odd about  $y_2$  to  $10^{-9}$ , equal to  $+2.27045$  on one side and  $-2.27045$  on the other, a jump of  $4.54089$  with  $C_+ = -C_- = 1.13522$ . The profile is therefore  $C^{1,1}$  but not  $C^2$ . It is not a logarithm: fitting  $f'' = 2D \log|y - y_2| + \text{const}$  returns a slope of  $0.043$  at  $R^2 = 0.46$ , consistent with zero. Correspondingly  $f'$  is continuous at both stagnation points, to  $4 \times 10^{-12}$  and  $1.6 \times 10^{-11}$ .

## 9 Determination of the global constant

Proposition 8 pins the stagnation point at which the profile is smooth, and which therefore contributes no singularity. The exponent  $\beta$  is set by the other one, whose  $c$ , written  $c_2$  below, is genuinely global: no local balance determines it.

Reading  $c_2$  off a single grid is unreliable. Across  $N = 512$  to  $4096$  it wanders between  $-1.6486$  and  $-1.6681$  without a monotone trend, and a three parameter fit of the form  $c_2(N) = c_2^{\infty} + AN^{-p}$  returns  $p = 3.3$  with amplitude  $1.1 \times 10^7$ , which is a fit to noise. The reason is the  $C^{1,1}$  regularity established above: every quantity read from the profile converges algebraically.

The remedy is that  $c_1$  is known exactly. Both constants are read from the same profile and carry the same discretisation error, entering linearly, so  $c_2$  is a linear function of  $c_1$  across resolutions and evaluating that line at the exact  $c_1$  cancels the error without requiring its size or its rate. At  $a = 0.8$  the regression over seven resolutions is

$$c_2 = -0.980361 c_1 + 3.240505, \quad R^2 = 0.99999744, \quad (15)$$

and the raw spread of  $1.95 \times 10^{-2}$  collapses to  $2.43 \times 10^{-5}$ , a factor of  $801$ . The corrected value from  $N = 512$  then lies within  $1.5 \times 10^{-5}$  of that from  $N = 4096$ . We obtain

$$c_2 = -1.66130027 \pm 1.0 \times 10^{-5}, \quad \mu_2 = 2.00242268, \quad \beta = 3.00242268. \quad (16)$$

**Remark 9** ( $\beta \neq 3$ ). The value  $\mu_2 = 2$ , and hence  $\beta = 3$ , would require  $c_2 = 1/(1 - 2a) = -1.66666667$  at  $a = 0.8$ . The corrected value misses this by  $5.37 \times 10^{-3}$ , some five hundred times the scatter of the corrected points. Since the correction removes eight hundred times the raw error, the conclusion survives allowing a residual systematic an order of magnitude above the observed scatter. Table 6 shows  $\beta$  varying continuously with  $a$ ; the value  $3$  is merely where that curve crosses.

| $a$  | $c_2$       | scatter               | $\mu_2$   | $\beta$   |
|------|-------------|-----------------------|-----------|-----------|
| 0.72 | -0.31519119 | $1.4 \times 10^{-14}$ | 5.7953854 | 6.7953854 |
| 0.74 | -0.56651088 | $4.2 \times 10^{-10}$ | 3.7367448 | 4.7367448 |
| 0.76 | -0.86560395 | $1.8 \times 10^{-8}$  | 2.8358720 | 3.8358720 |
| 0.78 | -1.22477889 | $1.3 \times 10^{-6}$  | 2.3288127 | 3.3288127 |
| 0.80 | -1.66130509 | $1.2 \times 10^{-5}$  | 2.0024205 | 3.0024205 |
| 0.82 | -2.20024724 | $5.3 \times 10^{-5}$  | 1.7737736 | 2.7737736 |

Table 6: The global constant and the resulting exponents. The scatter is the spread of the corrected values across resolutions and measures self consistency, not residual systematic error.

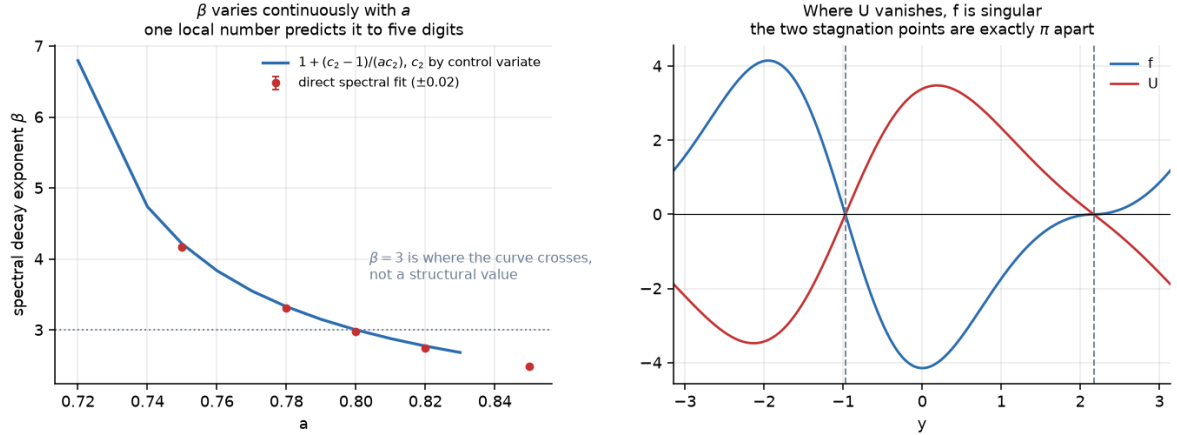


Figure 5: Left: the spectral decay exponent  $\beta$  against  $a$ , from (13) with  $c_2$  fixed by the control variate. The curve is smooth and  $\beta = 3$  is simply where it crosses. Direct fits of  $|\hat{f}_k| \sim k^{-\beta}$  to the spectrum are shown for contrast with error bars of  $\pm 0.02$ , which is their window sensitivity; the relation is four orders of magnitude more accurate than measuring the exponent directly. Right: the profile  $f$  and its velocity  $U$  at  $a = 0.8$ . The dashed lines mark the two zeros of  $U$ , which are exactly  $\pi$  apart, and  $f$  vanishes at both as Proposition 6 requires.

**Remark 10** (On measuring  $\beta$  from the spectrum directly). We caution against it. Fitting  $|\hat{f}_k| \sim k^{-\beta}$  over a band is sensitive to the band at the  $2 \times 10^{-2}$  level: shifting the window from  $128 \leq k \leq 512$  to  $0.094 k_{\text{cut}} \leq k \leq 0.375 k_{\text{cut}}$ , the same window to three digits, moves the answer from 3.00315 to 2.98626. Across ten reasonable windows a binned estimator spans 2.9829 to 3.0072 and an unbinned least squares fit spans 3.0132 to 3.0510, and the two barely overlap. The two ends of the spectrum are biased in opposite directions: the low end by the analytic factor multiplying  $|y - y_*|^\mu$ , which contributes  $k^{-(\mu+2)}$  and faster, and the high end by the dealiasing mask. Refinement distinguishes them, since the former is pinned to absolute  $k$  and the latter to a fixed fraction of  $k_{\text{cut}}$ . Equation (13) with a well determined  $c_2$  is four orders of magnitude more accurate than any of these fits.

## 10 A first integral and a global constraint

Equation (10) integrates in closed form. On any interval free of zeros of  $U$ ,

$$f = C |U|^{1/a} \exp\left(-\frac{1}{a} \int \frac{dy}{U}\right). \quad (17)$$

Near a simple zero this yields  $|y - y_*|^{1/a} |y - y_*|^{-1/(ac)} = |y - y_*|^\mu$ , recovering Proposition 6 independently.

**Proposition 11** (Global constraint). *If  $f$  and  $U$  are  $2\pi$  periodic and  $\log |f|$  has no jump at either zero of  $U$ , then*

$$\text{PV} \oint \frac{dy}{U} = 0. \quad (18)$$

*Proof.* Integrating (10) once around the circle,  $0 = \frac{1}{a} [0 - \text{PV} \oint dy/U]$ , since both  $\log |f|$  and  $\log |U|$  return to their initial values.  $\square$

The hypothesis holds here: at  $y_1$  the profile is smooth with  $\mu = 1$ , and at  $y_2$  the one sided constants satisfy  $C_+ = -C_-$ , so  $|C_+| = |C_-|$  and the logarithm does not jump. We verify (18) by subtracting both simple poles with  $\cot$ , which carries the same residue and whose own principal value on the circle vanishes. The resulting ratios are  $2.2 \times 10^{-12}$ ,  $6.7 \times 10^{-13}$ ,  $9.8 \times 10^{-13}$  and  $1.4 \times 10^{-12}$  at  $a = 0.75, 0.78, 0.80, 0.82$ , stable across three independent grid shifts.

**Remark 12.** A naive evaluation of (18) on the grid returns ratios between  $10^{-2}$  and 1 and appears to refute it. This is catastrophic cancellation: the stagnation points land on grid points to within  $10^{-11}$  of a cell, so the subtraction differences two quantities of size  $10^8$ . Evaluating on a half shifted grid, which is exact for a band limited field, resolves it.

**No closed form for  $c_2$ .** With  $c_2$  determined across eleven values of  $a$ , candidate shapes can be excluded rather than merely left unconfirmed. Every Möbius form is rejected:  $\mu_2 = (p + qa)/(1 + ra)$  leaves an rms residual of  $1.9 \times 10^{-4}$ ,  $c_2 = (p + qa)/(1 + ra)$  leaves  $3.0 \times 10^{-4}$ , and the same shape in the variable  $1/(1 - a)$  leaves  $6.0 \times 10^{-4}$ , each one to two orders above the data's precision. Fits of quadratic over quadratic achieve residuals of order  $10^{-7}$  but use five parameters against eleven points and dip below the data scatter at the rough end of the range, so they absorb smooth error rather than signal and we do not regard them as evidence.



We suggest this is the correct answer rather than a gap. Proposition 8 succeeds because it is a *local* balance at a point where  $f$  is smooth. No such balance is available at the second stagnation point, so its value is determined by the global solution. Writing  $\psi$  for the positive frequency part of  $U$ , so that  $U = \psi + \bar{\psi}$  and  $f = i(\psi' - \bar{\psi}')$ , equation (7) reads

$$a(\psi + \bar{\psi})(\psi'' - \bar{\psi}'') = (\psi' - \bar{\psi}')(\psi' + \bar{\psi}' - 1), \quad (19)$$

and the mixed products  $\psi\bar{\psi}''$  and  $\bar{\psi}\psi''$  straddle both halves of the spectrum. That failure to separate is exactly what makes  $a = 0$  integrable and  $a \neq 0$  not.

## 11 The critical limit $a \rightarrow 1$

The case  $a = 1$  is degenerate in a way no other member of the family is.

**Proposition 13** (Line of equilibria). *At  $a = 1$ , every  $A \sin x$  with  $A \in \mathbb{R}$  is a steady state of (2), as is every translate.*

*Proof.* With  $\omega = A \sin x$  we have  $\mathcal{H}\omega = -A \cos x$ ,  $u = -A \sin x$  and  $\omega_x = A \cos x$ , so  $\omega \mathcal{H}\omega = -A^2 \sin x \cos x = u \omega_x$  and  $\mathcal{R}_1(\omega) = \omega \mathcal{H}\omega - u \omega_x = 0$ .  $\square$

So the critical case carries a two parameter manifold of equilibria, amplitude and translation, which exists at  $a = 1$  and nowhere else. Writing  $\varepsilon = 1 - a$ , the family is destroyed at order  $\varepsilon$  and the blowup time diverges. We measure  $T$  without fitting: for a frozen profile  $\|\omega\|_\infty = \|f\|_\infty / (T - t)$  exactly, so  $T = t + \|f\|_\infty / \|\omega\|_\infty$  at any late time, and running to  $\|\omega\|_\infty = 10^6$  makes the correction term of order  $10^{-6}$ .

| $a$            | 0.80   | 0.90    | 0.93    | 0.95    | 0.96    | 0.97    | 0.98    | 0.99    |
|----------------|--------|---------|---------|---------|---------|---------|---------|---------|
| $T$            | 6.0719 | 10.7569 | 14.0844 | 17.7129 | 20.4591 | 24.5830 | 32.1786 | 56.9742 |
| $T\varepsilon$ | 1.214  | 1.076   | 0.986   | 0.886   | 0.818   | 0.737   | 0.644   | 0.570   |
| local $p$      | -      | 0.828   | 0.755   | 0.681   | 0.646   | 0.638   | 0.664   | 0.824   |

Table 7: Blowup time as  $a \rightarrow 1$ , and the local exponent in  $T \sim \varepsilon^{-p}$  between consecutive entries.  $T\varepsilon$  decreases throughout, so  $T$  grows strictly slower than  $1/\varepsilon$ , but  $p$  does not converge.

The exponent does not settle: it falls from 0.83 to 0.64 and returns to 0.82 at the smallest  $\varepsilon$ . No functional form fits the last points,  $C\varepsilon^{-p}$ ,  $C/\varepsilon + D$ ,  $C \log(1/\varepsilon) + D$  and  $A/\varepsilon + B \log(1/\varepsilon) + C$  leaving rms residuals of 0.99, 0.71, 4.22 and 0.38 against  $T$  values from 6 to 57. We therefore quote no exponent. What follows instead is an account of why the natural reductions do not settle it either.

### 11.1 The two mode reduction is exact, and gives the wrong answer

Expanding about the manifold, put  $\omega = A \sin x + B \sin 2x$ . Using  $\mathcal{H} \sin kx = -\cos kx$  and  $u = -k^{-1} \sin kx$ ,

$$\begin{aligned} \omega \mathcal{H}\omega &= -\frac{A^2}{2} \sin 2x - AB \sin 3x - \frac{B^2}{2} \sin 4x, \\ u \omega_x &= -\frac{A^2}{2} \sin 2x - \frac{5AB}{4} \sin 3x + \frac{3AB}{4} \sin x - \frac{B^2}{2} \sin 4x, \end{aligned}$$

so that  $\mathcal{R}_a = \omega \mathcal{H}\omega - a u \omega_x$  has coefficients  $-\frac{3a}{4}AB$  on  $\sin x$ ,  $-\frac{\varepsilon}{2}A^2$  on  $\sin 2x$ ,  $(\frac{5a}{4} - 1)AB$  on  $\sin 3x$  and  $-\frac{\varepsilon}{2}B^2$  on  $\sin 4x$ , each verified against the solver to  $10^{-17}$ .

The  $\sin x$  terms cancel in the first line but not the second, so the feedback of mode two into mode one survives at *linear* order. Any argument predicting  $p = 2/3$  from a parity cancellation is therefore excluded, not by numerics but by the calculation.

Retaining the two modes gives  $A' = -\frac{3a}{4}AB$ ,  $B' = -\frac{\varepsilon}{2}A^2$ , which integrates in closed form. Setting  $L = \log A$  at  $a = 1$  gives  $L' = -\frac{3}{4}B$  and  $L'' = \frac{3\varepsilon}{8}e^{2L}$ , a Liouville equation; multiplying by  $L'$  and using  $L(0) = L'(0) = 0$  gives  $L'^2 = \frac{3\varepsilon}{8}(e^{2L} - 1)$ , and since  $\int_0^\infty (e^{2L} - 1)^{-1/2} dL = \arccos(e^{-L})|_0^\infty = \pi/2$ ,

$$T_2 = \frac{\pi}{2} \sqrt{\frac{8}{3\varepsilon}} = 2.56510 \varepsilon^{-1/2}. \quad (20)$$

Against a two mode Galerkin truncation the ratio  $T/T_2$  reads 1.118, 1.054, 1.026, 1.010, 1.005 at  $\varepsilon = 0.2, 0.1, 0.05, 0.02, 0.01$ . The reduction is exact and gives  $p = 1/2$ , which is not the behaviour of the full equation.

## 11.2 The exponent is carried by the cascade

Retaining  $m$  modes and measuring the exponent over  $0.02 \leq \varepsilon \leq 0.1$  gives the following.

| $m$ | 2     | 4     | 6     | 8     | 12    | 24    |
|-----|-------|-------|-------|-------|-------|-------|
| $p$ | 0.474 | 0.506 | 0.518 | 0.525 | 0.536 | 0.554 |

Table 8: Exponent from an  $m$  mode Galerkin truncation. The value climbs monotonically and is still climbing at  $m = 24$ ; at  $\varepsilon = 0.1$  the gap to the full answer closes like  $m^{-1/2}$ .

No finite truncation reproduces the exponent. The case  $m = 3$  is degenerate and sits outside the trend, giving  $T = 405$  at  $\varepsilon = 0.1$  and no blowup at all at 0.05 or 0.02.

## 11.3 There is no spectral gap, hence no reduction

Both failures have one cause. Since  $\mathcal{R}$  is quadratic,  $D\mathcal{R}_1(A \sin x) = A M$  with  $M = D\mathcal{R}_1(\sin x)$  independent of  $A$ , and writing  $\omega = A \sin x + v$ ,

$$v_t = A M v - \frac{\varepsilon}{2} A^2 \sin 2x, \quad A' = -\frac{3}{4} A b_2, \quad (21)$$

with  $b_2$  the  $\sin 2x$  component of  $v$ . Were  $M$  strictly stable,  $v$  would saturate at  $\frac{\varepsilon}{2} A M^{-1} \sin 2x$  and (21) would reduce to  $A' = \kappa \varepsilon A^2$ , a Riccati equation giving  $p = 1$ . That step is unavailable.

**Observation 14** (No gap). The spectrum of  $M$  is purely imaginary. Computed densely at  $N = 128, 256, 512$ , every eigenvalue has real part at the  $10^{-5}$  roundoff level against imaginary parts reaching 72, and at  $N = 256$  none has positive real part. Its kernel contains the tangent directions to the manifold exactly:  $\|M \sin x\|_\infty = 8 \times 10^{-14}$  and  $\|M \cos x\|_\infty = 6 \times 10^{-14}$ .

Moreover  $M$  is not diagonalisable. Integrating  $v_t = M v$  from a random perturbation,  $\|v\|$  grows linearly, by factors 187, 736, 1392, 2873, 5738 at  $t = 10, 50, 100, 200, 400$ , doubling as  $t$  doubles; with imaginary spectrum this means a Jordan block at zero. Splitting the state identifies its origin. The components along  $\sin x$  and  $\cos x$  grow linearly, reaching 2663 and

$-845$  at  $t = 400$ , while the remainder stays bounded and oscillates between 135 and 467. The secular growth is therefore *drift along the manifold of equilibria* rather than instability: a perturbation alters the amplitude and phase at a constant rate, carrying the state to a different steady state rather than away from all of them. The Jordan block is generated by the symmetries themselves, consistent with the stability theorem of Jia, Stewart and Sverak [4], which is stability modulo exactly these symmetries.

We note also that  $M$  is strongly non-normal: the remainder above is amplified from 7.8 to 467 before returning, a transient factor near 60 with no eigenvalue to account for it.

The consequence is structural. Modulo the kernel every mode oscillates without decay and the kernel modes drift linearly, so there is no gap anywhere. Hence quasi steady elimination is invalid, because the transient it discards never dies; mode truncation is not a reduction, because the discarded modes never relax, which is why the truncated exponent converges only algebraically; and there is no centre manifold to reduce onto, because the centre subspace is the whole space. The cascade cannot be eliminated because nothing within it is fast relative to anything else, and (20), exact yet giving the wrong exponent, is the sharpest available demonstration that a finite reduction misses the mechanism rather than approximating it.

## 12 Limitations

The datum  $\sin x$  is not generic: it is precisely the ground state of the  $a = 1$  model. Table 4 and Section 6 between them answer the worst form of this objection, reaching the same profile from sixteen random data, from a continuous family of offsets and from three designed data, always with the same constants to seven digits.

What remains open is the shape of the basin boundary rather than its existence. Two one parameter families were probed in a space of functions, and the decay window of Section 6 already shows the boundary is not simply whether the datum changes sign. Nothing here maps it in more than one dimension, and the interior of the decay window is not resolved:  $c = 0.52$  goes under resolved at  $t = 42$  while still decaying.

All statements about  $a = 0.9$  and above are unresolved. The residual of the profile equation there does not fall monotonically with the amplitude cap, taking values 0.78, 2.93, 1.77 at caps  $10^4$ ,  $10^6$ ,  $10^8$ , so those measurements reflect the fitting window rather than the solution. Since  $T(0.9) \approx 10.8$  against 6.1 at  $a = 0.8$ , comparisons should be made at a fixed fraction of  $T$  rather than a fixed amplitude.

The scatter quoted in Table 6 measures how consistently the control variate reproduces itself across resolutions. A residual nonlinear dependence of the discretisation error on  $c_1$  would cancel from it and go undetected, so the figure of  $1.4 \times 10^{-14}$  at  $a = 0.72$  should not be read as fourteen digit accuracy.

On priority. The bibliographic details of every reference above have been verified. Seven closely related works, [3], [11], [12], [13], [14], [15] and [16], have been read for overlap and none contains Propositions 6 or 8, the stability computation, or the regularity statement. We note that this list grew twice while being compiled, so it should be read as the nearest work we could find rather than as a complete survey. This is a narrower check than a full literature survey: it covers the papers nearest to this one rather than the field, and for the two preprints it rests on a reading of the full text rather than a line by line verification of every lemma. Both numerical coincidences with [16] have been examined and neither is an overlap, but the first

of them, the exchange of stability near  $a \approx 0.75$ , is stated there as a hypothesis and would be worth settling on its own merits.

The critical limit is left open. Section 11 reports no exponent for  $T(\varepsilon)$  because the measured one does not converge over the range we can reach, and reaching  $\varepsilon = 10^{-3}$  would need some thirty times the 866,000 steps that  $\varepsilon = 10^{-2}$  already took. Observation 14 explains why the shortfall cannot be made up by analysis in place of computation, but it is an explanation of a difficulty rather than a resolution of it.

Finally, nothing here is a proof. A computer assisted proof in this setting, in the style of Chen and Hou [9], requires interval arithmetic producing certified enclosures rather than floating point with convergence studies. The objects this paper produces, a profile equation with a linearly stable solution and an exact relation fixing one of its two local constants, are the objects such a proof would need to enclose.

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